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Using Continuous $PM_{2.5}$ Monitoring Data to Report an Air Quality Index

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ABSTRACT

As stated in 40 CFR 58, Appendix G (2000), statistical linear regression models can be applied to relate $PM_{2.5}$ continuous monitoring (CM) measurements with federal reference method (FRM) measurements, collocated or otherwise, for the purpose of reporting the air quality index (AQI). The CM measurements can then be transformed via the model to remove any bias relative to FRM measurements. The resulting FRM-like modeled measurements may be used to provide more timely reporting of a metropolitan statistical area's (MSA's) AQI.[1] Of considerable importance is the quality of the model used to relate the CM and FRM measurements. The use of a poor model could result in misleading AQI reporting in the form of incorrectly claiming either good or bad air quality.

This paper describes a measure of adequacy for deciding whether a statistical linear regression model that relates FRM and continuous $PM_{2.5}$ measurements is sufficient for use in AQI reporting. The approach is the U.S. Environmental Protection Agency's (EPA's) data quality objectives (DQO) process, a seven-step strategic planning approach to determine the most appropriate data type, quality, quantity, and synthesis for a given activity.[2] The chosen measure of model adequacy is r^2, the square of the correlation coefficient between FRM measurements and their modeled counterparts. The paper concludes by developing regression models that meet this desired level of adequacy for the MSAs of Greensboro/Winston-Salem/High Point, NC; and Davenport/Moline/Rock Island, IA/IL. In both cases, a log transformation of the data appeared most appropriate. For the data from the Greensboro/Winston-Salem/High Point MSA, a simple linear regression model of the FRM and CM measurements had an r^2 of 0.96, based on 227 paired observations. For the data from the Davenport/Moline/Rock Island MSA, due to seasonal differences between CM and FRM measurements, the simple linear regression model had to be expanded to include a temperature dependency, resulting in an r^2 of 0.86, based on 214 paired observations.

IMPLICATIONS

According to 40 CFR 58.50 (2000), all MSAs with a population of at least 350,000 are required to report daily air quality, using the AQI, to the general public.[3] The AQI is calculated in part from concentrations of particulate matter (PM). Because $PM_{2.5}$ FRMs often report data less frequently than daily, continuous $PM_{2.5}$ monitors may be used to provide more timely AQI reporting, provided a relationship between the continuous and FRM measurements is established by statistical linear regression.[3] This paper describes a measure of adequacy for such a relationship and develops regression models for two case studies. These models may be useful for other applications requiring daily estimates of FRM-like $PM_{2.5}$ air quality.

INTRODUCTION

The air quality index (AQI) is an index for reporting to the public information about the local air quality and whether air pollution levels pose a health concern. It is a national index, meaning that the values and associated level of health concern are the same throughout the United States. The AQI focuses on health effects that can happen within a few hours or days and is calculated from the concentrations of five criteria pollutants: O_3, particulate matter (PM), CO, SO_2, and NO_2.

Regarding PM, the current $PM_{2.5}$ federal reference method (FRM) monitoring network is comprised of filter-based samplers. The definition and operation of $PM_{2.5}$ FRMs is explicitly defined in 40 CFR Part 50 (1997).[4] These samplers provide measurements that are consistent and repeatable both temporally and geographically.[5] However, the measurements are not provided in a manner that is timely enough for use in the reporting of the AQI. Weeks can pass before the concentrations for a day are reported. According to 40 CFR 58, Appendix G (2000), PM measurements

from non-federal reference or federal equivalent method monitors (non-FRM/FEM) may be used for the purpose of reporting the AQI.[3] For example, non-FRM/FEM PM methods include tapered element oscillating microbalances (TEOM), nephelometers, beta attenuation monitors (BAM), and continuous aerosol mass monitors (CAMM). The primary advantage of these non-FRM/FEM methods is their timely, time-resolved, cost-efficient measurements. However, due to the different method of measuring PM, their concentrations may differ from those of an FRM/FEM. For example, typical operating procedures for the TEOM and BAM involve heating the inlet to 30 or 50 ºC to reduce the amount of particle-bound water. Such heating may bias the measurement when there is volatile PM present. Because of these potentially different estimates of PM, CFR requires that non-FRM/FEM measurements be used for AQI reporting only after a statistical linear relationship has been established with an FRM/FEM.

Several studies have investigated relationships between measurements from different types of PM samplers, and one recent study compares measurements between FRM $PM_{2.5}$ monitors and non-FRM monitors.[6-11] Conclusions from these studies indicate that certain types of monitors compare well, while others do not. The conclusion about whether a comparison is good or bad is typically based on a qualitative assessment of the slope of the regression, the intercept of the regression, and the correlation coefficient. People may draw different conclusions about whether a comparison is good or bad because the decision is subjective.

This paper describes a quantitative measure of adequacy for statistical linear regression models relating FRM and continuous $PM_{2.5}$ measurements for the purpose of reporting the AQI. To guide appropriate model development of the relationship between FRM and continuous monitoring (CM) measurements, we make use of the data quality objectives (DQO) process, a seven-step strategic planning approach, based on the scientific method, for determining the appropriate data type, quality, quantity, and synthesis.[2] The decision about whether a relationship developed between an FRM and continuous $PM_{2.5}$ monitor is adequate depends on the number of paired samples, the correlation, and the size of the errors considered acceptable.

Next, this paper provides two case studies in which models that meet the specified level of adequacy are developed. The case studies include the metropolitan statistical areas (MSAs) of Greensboro/Winston-Salem/High Point, NC, henceforth referred to as NC; and Davenport/Moline/Rock Island, IA/IL, henceforth referred to as IA/IL. In both case studies, the FRM was a Partisol-Plus 2025 sequential sampler, manufactured by Rupprecht & Patashnick Co., and the non-FRM was a TEOM with a size selective inlet to remove particles larger than 2.5 µm. The TEOM in NC was continuously operated at 50 ºC, and the TEOM in IA/IL was operated at 50 ºC during the summer and 30 ºC during the winter.

THE DQO PROCESS

The DQO process is a planning tool to clarify and document the specific use of environmental data for making decisions and to ensure efficiency in collecting data that meets the needs of decision-makers.[2] It divides the process into seven steps: state the problem, identify the decision, identify the inputs to the decision, define the boundaries of the study, develop a decision rule, specify tolerable limits on decision errors, and optimize the design for obtaining data. Briefly, each of these seven parts and the activities associated with them is considered in the sections that follow. The process, however, is not entirely linear. Some iteration between steps is often required because the results of the later steps may impact earlier ones. Ideally, the last three steps of the DQO process should be done in parallel.

State the Problem (Step 1)

The purpose of this step is to concisely define the problem at hand. *Problem Statement:* It is desired to use continuous $PM_{2.5}$ measurements for the purpose of reporting an AQI in, for example, NC or IA/IL. According to 40 CFR 58, Appendix G (2000), these data may be used for this purpose if a linear relationship between continuous measurements and reference or equivalent $PM_{2.5}$ method measurements can be established by statistical linear regression. Therefore, a model relating FRM and continuous $PM_{2.5}$ measurements, possibly adjusting for meteorological data, is required.

Identify the Decision (Step 2)

The purpose of this step is to clearly define the decision statement the study will attempt to resolve. *Decision Statement:* Is the statistical linear model relating FRM $PM_{2.5}$ measurements to continuous $PM_{2.5}$ measurements acceptable for transforming continuous measurements for the purpose of reporting, for example, the NC or IA/IL AQI? If so, then the continuous $PM_{2.5}$ data, along with the model, can be used to report the AQI. If not, continuous $PM_{2.5}$ data should not be used to report the AQI.

Identify the Inputs to the Decision (Step 3)

The purpose of this step is to identify the informational inputs needed to resolve the decision statement and determine the inputs that require environmental measurements. The environmental measurements required for this study are FRM $PM_{2.5}$ daily measurements, continuous $PM_{2.5}$ hourly measurements, and, possibly, meteorological data

such as temperature. At the most basic level, an MSA will require a set of days for which both FRM $PM_{2.5}$ measurements and CM $PM_{2.5}$ measurements have been obtained from sites within the MSA. Such information is vital to developing a model relating the two measures. Ideally, (1) a large number of days will be available, including data spanning at least one year; (2) at least some of the FRM and CM data will be collocated; and (3) meteorological data will be available for model improvement, if necessary. For example, both NC and IA/IL have monitoring sites producing collocated FRM and CM measurements. In many cases, these data will be available in EPA's aeromatic information retrieval system (AIRS) database. In some cases, data will be accessible from an MSA's own data archive.

Define the Boundaries of the Study (Step 4)

The purpose of this step is to define the spatial and temporal boundaries covered by the decision statement. The population of interest is daily FRM/FEM $PM_{2.5}$ concentrations for the MSA, measured in $\mu g/m^3$. The MSA is the geographical area within which the decision statement is to be applied. Data permitting, some MSAs might develop models specific to subregions within the MSA; hence, the spatial scale of decision-making could be anywhere from an MSA subregion surrounding the site(s) used to develop the model to the entire MSA itself.

The time frame to which the decision applies will be up to individual MSA decision-makers. It is recommended that an acceptable model should be checked for accuracy and updated, if necessary, at least yearly, if not quarterly or even more frequently. Hence, the temporal bounds of decision-making might range from a few days, if a model is continuously updated or replaced, to an entire year, if the MSA decision-makers feel the model is still accurate a year after development.

Develop a Decision Rule (Step 5)

The purpose of this step is to define the parameter of interest that quantifies the decision, specify the action level, and integrate previous DQO outputs into a single statement that describes a logical basis for choosing among alternative actions. Because the overall purpose of this exercise is to develop an acceptable model that relates FRM and CM measurements, it was determined that the parameter of interest is R^2. In general, R^2 measures the strength of the model fit to the data. In this case, R^2 is interpreted as the correlation between measured and modeled FRM data.

The action level around which a model will be deemed acceptable was determined to be $R^2 = 0.60$. At first, this action level might appear somewhat lax to data analysts used to interpreting strong regression relationships as those with an observed r^2 value in the range of 0.80 or greater. However, it is important to keep in mind that, in the current context, a decision is to be made based on estimating the model's true R^2 value, a rather uncommon activity in practice. In most applied contexts, the sample statistic r^2 obtained from software regression output is interpreted as the true R^2 value, when in fact it is only an estimate of the true unknown value. Under a hypothesis testing scenario, accepting or rejecting a model based on a true R^2 action level of 0.60 will be seen to be equivalent to requiring a sample r^2 value equal to ~0.80, a model adequacy threshold more common to most applied data analysts. *"If…then…" Statement:* If the true R^2 value of the statistical linear regression model relating FRM and continuous $PM_{2.5}$ measurements within, for example, NC or IA/IL is greater than 0.60, then continuous $PM_{2.5}$ data can be used, along with the model, to report the AQI for the next 90-day to 1-year period. Otherwise, the model in its current form is not acceptable and should not be used for AQI reporting.

Specify Tolerable Limits on Decision Errors (Step 6)

The purpose of this step is to specify tolerable limits on decision errors. The decision as to whether the model is acceptable may be statistically formalized as the following hypothesis test:

$$H_0: R^2 \leq 0.60 \text{ vs. } H_a: R^2 > 0.60 \quad (1)$$

where R (and, hence, R^2) can theoretically range from 0.0 (i.e., no relation between actual and modeled FRM $PM_{2.5}$ measurements) to 1.0 (i.e., perfect correlation between actual and modeled FRM $PM_{2.5}$ measurements). The null, or baseline, hypothesis of $R^2 \leq 0.60$ is chosen because the decision error associated with this conclusion is considered to be the most serious and, thus, should be guarded against. Specifically, the false rejection decision error that the model is adequate ($R^2 > 0.60$) when, in fact, it is not ($R^2 \leq 0.60$) could result in misleading AQI reporting in the form of incorrectly claiming either good or bad air quality. In contrast, the false acceptance decision error that the model is unsatisfactory ($R^2 \leq 0.60$) when, in fact, it is adequate ($R^2 > 0.60$) simply results in not using the model in its current form and developing a model with a better correlation coefficient.

Along with the above hypothesis statement, three additional parameters must be specified to formally accept or reject the model; namely, the false rejection decision error rate (α), the false acceptance decision error rate (β), and the size of the "gray region" in decision-making (Δ). The false rejection decision error rate (α) specifies the maximum probability of claiming the model is adequate

($R^2 > 0.60$) when, in fact, it is not ($R^2 \leq 0.60$). The false acceptance decision error rate (β) specifies the maximum probability of claiming the model is not adequate ($R^2 \leq 0.60$) when, in fact, it is ($R^2 > 0.60$). The size of the gray region in decision-making (Δ) specifies an area, starting at $R^2 = 0.60$ up to $R^2 = (0.60 + \Delta)$, within which somewhat higher false acceptance decision error rates (β) are considered tolerable. For the purposes of this paper, values of $\alpha = 0.05$, $\beta = 0.30$, and $\Delta = 0.25$ will be used.

Figure 1 provides a visual interpretation of the meaning of each of the three chosen DQO parameters ($\alpha = 0.05$, $\beta = 0.30$, and $\Delta = 0.25$). The figure draws a curve indicating the probability of claiming the true R^2 value is above the action level of 0.60 (vertical axis) as a function of the true unknown R^2 value (horizontal axis). Notice that for all values of $R^2 \leq 0.60$, the curve remains below the 0.05 threshold on the vertical axis. In other words, if the model is truly unacceptable ($R^2 \leq 0.60$), then the chance of claiming otherwise is never more than 5% (i.e., $\alpha = 0.05$). Likewise, if the model is quite good ($R^2 \geq 0.85$), then the chance of claiming otherwise is never more than 30% (i.e., $\beta = 0.30$). Finally, if the model is good, but only marginally so ($0.60 < R^2 \leq 0.85$), then the chance of claiming otherwise could be substantial (i.e., more than 30%).

Optimize the Design for Obtaining Data (Step 7)

The purpose of this step is to identify a resource-effective data collection design for generating data that are expected to satisfy the DQOs. In the current context, the nature of the data collection design is predominantly determined by the existing infrastructure that an MSA, such as NC or IA/IL, has in place to develop a model. How many FRM or CM monitoring station sites have been collecting data? Are there collocated FRM and CM measurements, as in the cases of NC and IA/IL? Which sites should be used in developing a model (e.g., collocated core FRM and CM sites only)? Assuming these questions are answered, the major design issue remaining is the minimum number of days required for adequate model development. The following provides the statistical rationale for determining this sample size requirement. Along the way to getting a sample size, the practical rule for deciding whether the model is good enough is developed.

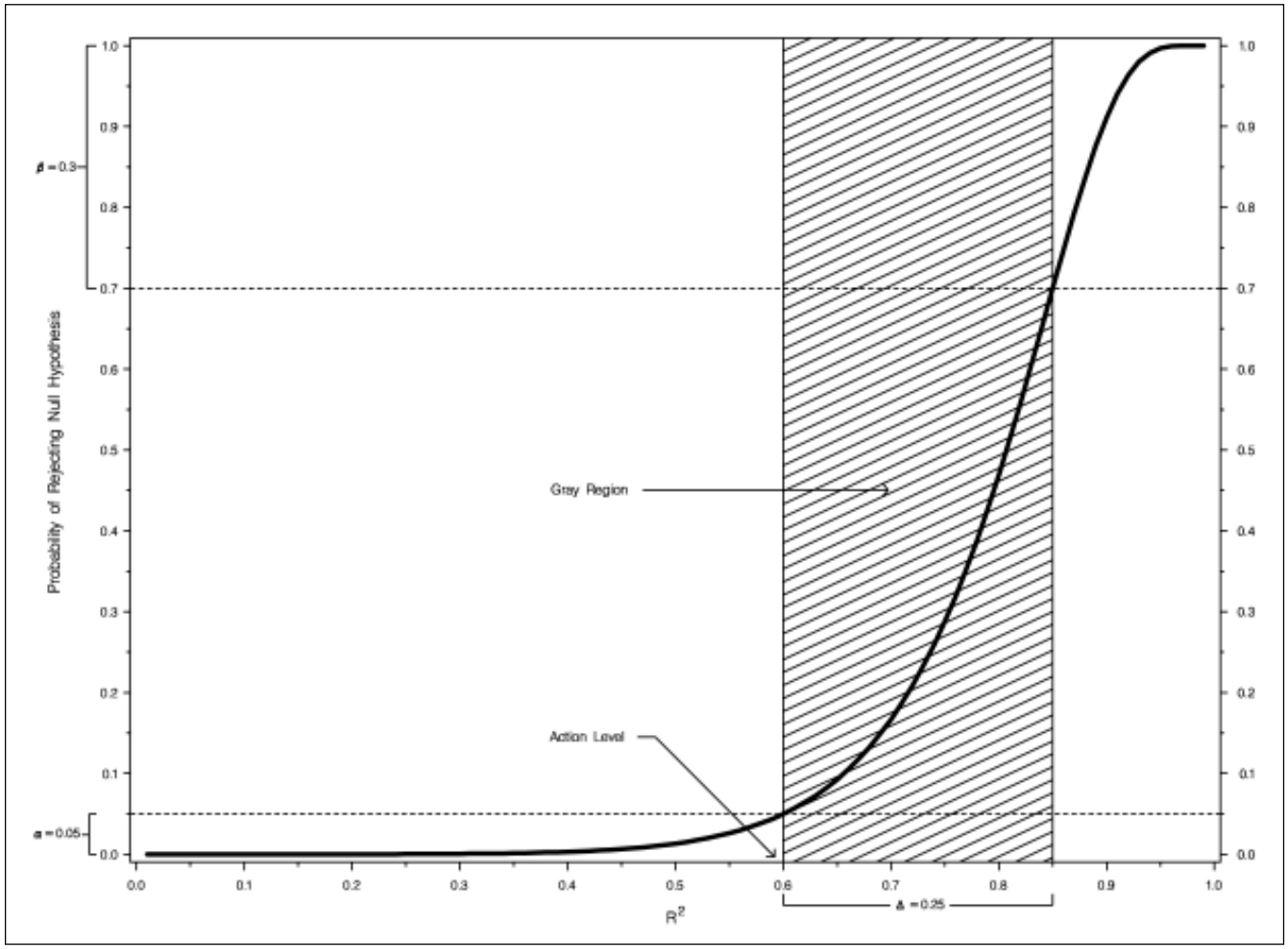


Figure 1. Example power curve when $\alpha = 0.05$, $\beta = 0.30$, and $\Delta = 0.25$.

First, define the parameter W as follows:

$$W = \frac{1}{2}\ln\left(\frac{1+R}{1-R}\right) \tag{2}$$

Assuming the observed FRM $PM_{2.5}$ data values and their associated predictions from the model follow a bivariate normal distribution, it follows that W has an approximate normal distribution with mean $^1/_2 \ln [(1 + R)/(1 – R)]$ and variance $^1/\sqrt{n-3}$, where R equals the square root of the true unknown R^2 value.[12] Testing the null hypothesis specified in eq 2 is thus equivalent to a test of

$$H_0:\ W \le \frac{1}{2}\ln\left(\frac{1+\sqrt{0.60}}{1-\sqrt{0.60}}\right) = 1.0317 \tag{3}$$

To conduct an α-level hypothesis test, we require

$$P_{R^2=0.60}\{W > c\} = \alpha \tag{4}$$

where the value c is chosen so as to satisfy the above equality. In words, eq 4 implies that the probability of incorrectly concluding the model is acceptable is limited to no more than α. Under the assumption of normality, eq 4 is equivalent to

$$P\left\{Z > (c - 1.0317)(n-3)^{1/4}\right\} = \alpha \tag{5}$$

where W has been transformed to Z (a standard normal random variable with a mean of 0 and a variance of 1) by subtracting off its mean when $R^2 = 0.60$ and dividing by its standard deviation. Based on the distribution of Z, eq 5 is satisfied when

$$(c - 1.0317)(n-3)^{1/4} = z_\alpha \tag{6}$$

where z_α is the αth percentile of the standard normal distribution. Solving for c gives

$$c = \frac{z_\alpha}{(n-3)^{1/4}} + 1.3017 \tag{7}$$

Next, to obtain sample size requirements, we consider the DQO parameters β and Δ. Our requirement is

$$P_{R^2 \ge 0.60+\Delta}\{W > c\} \ge 1 - \beta \tag{8}$$

This is equivalent to

$$P\left\{Z > \left[c - \frac{1}{2}\ln\left(\frac{1+\sqrt{0.60+\Delta}}{1-\sqrt{0.60+\Delta}}\right)\right](n-3)^{1/4}\right\} \ge 1 - \beta \tag{9}$$

where, as above, W has been transformed to Z by subtracting its mean and dividing by its standard deviation. Based on the distribution of Z, eq 9 is satisfied when

$$\left[c - \frac{1}{2}\ln\left(\frac{1+\sqrt{0.60+\Delta}}{1-\sqrt{0.60+\Delta}}\right)\right](n-3)^{1/4} \le z_{1-\beta} \tag{10}$$

where $z_{1-\beta}$ is the (1–β)th percentile of the standard normal distribution. Substituting in for c and solving for n gives the formula for calculating the required model development sample size as

$$n \ge \left[\frac{z_\alpha - z_{1-\beta}}{\frac{1}{2}\ln\left(\frac{1+\sqrt{0.60+\Delta}}{1-\sqrt{0.60+\Delta}}\right) - 1.0317}\right]^4 + 3 \tag{11}$$

Finally, because c is the point at which the model is determined to be acceptable on the scale of W, transforming c back to the scale of r^2 gives the formula for calculating the observed r^2 model acceptance lower bound as

$$r^2 \ge \left[\frac{\exp(2c)-1}{\exp(2c)+1}\right]^2 \tag{12}$$

where, given the specification of n in eq 11, c is as defined in eq 7. Using the DQO parameters specified in Step 6 ($\alpha = 0.05$, $\beta = 0.30$, and $\Delta = 0.25$), eqs 11 and 12 dictate that at least 213 days of data be used for model development and the resulting r^2 value be at least 0.81 for the model to be considered acceptable for its intended use.

Note that smaller values of α, β, and Δ demand more data for establishing a model and that, generally, the model can have a lower observed r^2. For example, if $\alpha = 0.05$, $\beta = 0.20$, and $\Delta = 0.25$, then a model is considered to be adequate if it is based on at least 365 data points and has an observed r^2 of 0.79 or greater. Similarly, larger values of α, β, and Δ demand less data to establish a model and, generally, the model must have a larger correlation coefficient. For example, if $\alpha = 0.10$, $\beta = 0.30$, and $\Delta = 0.25$, then a model is considered to be adequate if it is based on at least 104 data points and the observed r^2 is 0.80 or greater.

RESULTS

To relate CM data with FRM data, hourly CM measurements were averaged to produce a daily value. Consistent with 40 CFR Part 50, Appendix H (2000), a criterion of 75% completeness, 18 of 24 hr, was used to determine valid CM daily averages (specified in DQO Step 3).[5] The data were all well above the method detection limit. For this paper, we present only the results for the collocated CM and FRM data of NC and IA/IL. Figure 2 displays scatterplots of FRM measurements versus daily average CM measurements for NC and IA/IL, with a 1-1 line overlaid. A bias between the two measurements will show up

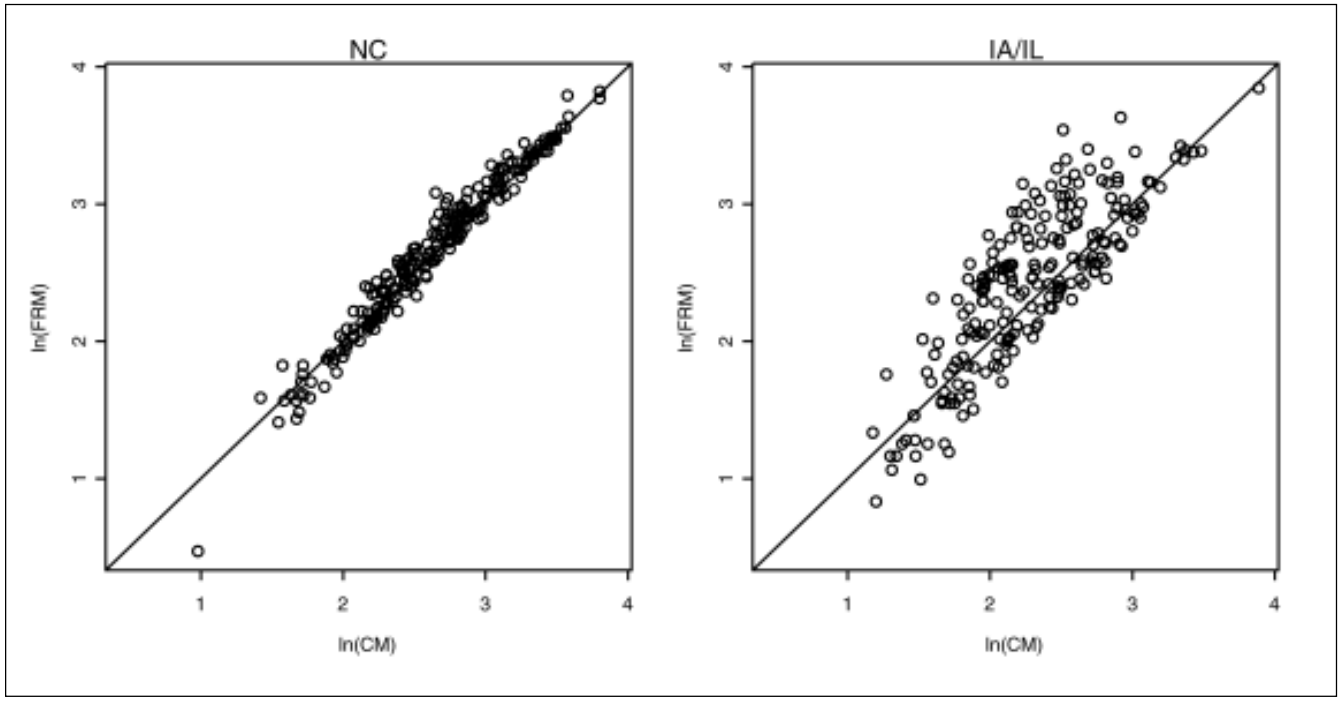


Figure 2. Scatterplots of FRM measurements vs. CM measurements in NC and IA/IL, on the natural log scale.

as a clustering of the data away from the 1-1 line, a phenomenon slightly more noticeable for the IA/IL data.

Model Development

In both cases, NC and IA/IL, the data were first transformed via the natural log function to better satisfy the normality assumption of DQO Step 7. The correlation between FRM and CM data was very strong in NC, requiring only a simple log-linear regression model {$\ln(\text{FRM}) = \beta_0 + [\beta_1 \times \ln(\text{CM})]$ + residual error} to satisfy the model quality requirements of DQO Step 7. The IA/IL data, however, required an additional temperature term in the log-linear model {$\ln(\text{FRM}) = \beta_0 + [\beta_1 \times \ln(\text{CM})] + (\beta_2 \times \text{temperature})$ + residual error} to account for seasonal differences between FRM and CM measurements.

Specifically, the simple log-linear regression model in IA/IL yielded an r^2 of 0.69, whereas adding daily average temperature to the model improved r^2 to 0.86. The seasonal differences observed in IA/IL may have been due to the fact that the TEOM in IA/IL was operated at 50 ºC during the summer and 30 ºC during the winter. One model adjustment that was considered to account for this observed effect in the data was to include an indicator, or mean shift variable, for season (i.e., mode of operation). However, including temperature as a continuous covariate proved to be a superior model adjustment.

Figure 3 demonstrates why a slightly more complicated model was required in IA/IL. For both the NC and the IA/IL data, the figure plots the time series of the difference between FRM and CM measurements on the natural log scale, with a smooth trend overlaid. (The smooth trend was estimated with the loess function in S-Plus.[13]) The NC time series of FRM-CM differences exhibits no clear seasonal pattern, whereas the IA/IL differences oscillate between positive and negative values, depending on season. A plausible explanation for this seasonal pattern is the effect of temperature on one or both measuring devices. Hence, a temperature term was added to the IA/IL model to adjust for the seasonally dependent differences between FRM and CM measurements in this case.

Table 1 provides regression summary statistics for the final models developed in the two case studies. Notice the very strong relationship, $r^2 = 0.96$, between FRM and

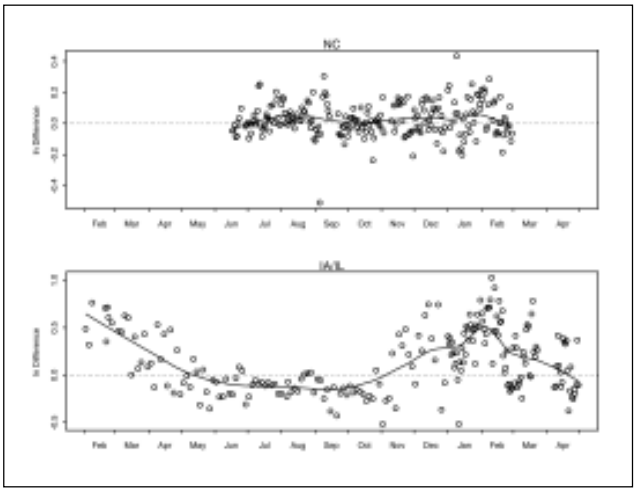


Figure 3. Time series of [ln(FRM) – ln(CM)] for NC (top) and IA/IL (bottom).

Table 1. Regression summary statistics for final models in NC and IA/IL.

MSA	Days of Data: N	Intercept: β_0 (s.e.)	ln(CM) Effect: β_1 (s.e.)	Temp. Effect: β_2 (s.e.)	RMSE	r^2
NC	227	–0.114 (0.036)	1.054 (0.013)	NA	0.104	0.96
IA/IL	214	–0.269 (0.078)	1.280 (0.036)	–0.027 (0.002)	0.225	0.86

CM daily values in NC. In both cases, NC and IA/IL, the sample sizes and r^2 values were considered sufficient evidence, according to DQO Step 7, that the models are appropriate for use in reporting the AQI.

Figure 4 provides further regression diagnostic summaries. The residual histograms in Figure 4 indicate that the data indeed are approximately normally distributed after being transformed via the natural log function. The lack of any discernible trends in the residual-versus-predicted scatterplots of Figure 4 provides evidence that the chosen NC and IA/IL models are appropriate for their respective data.

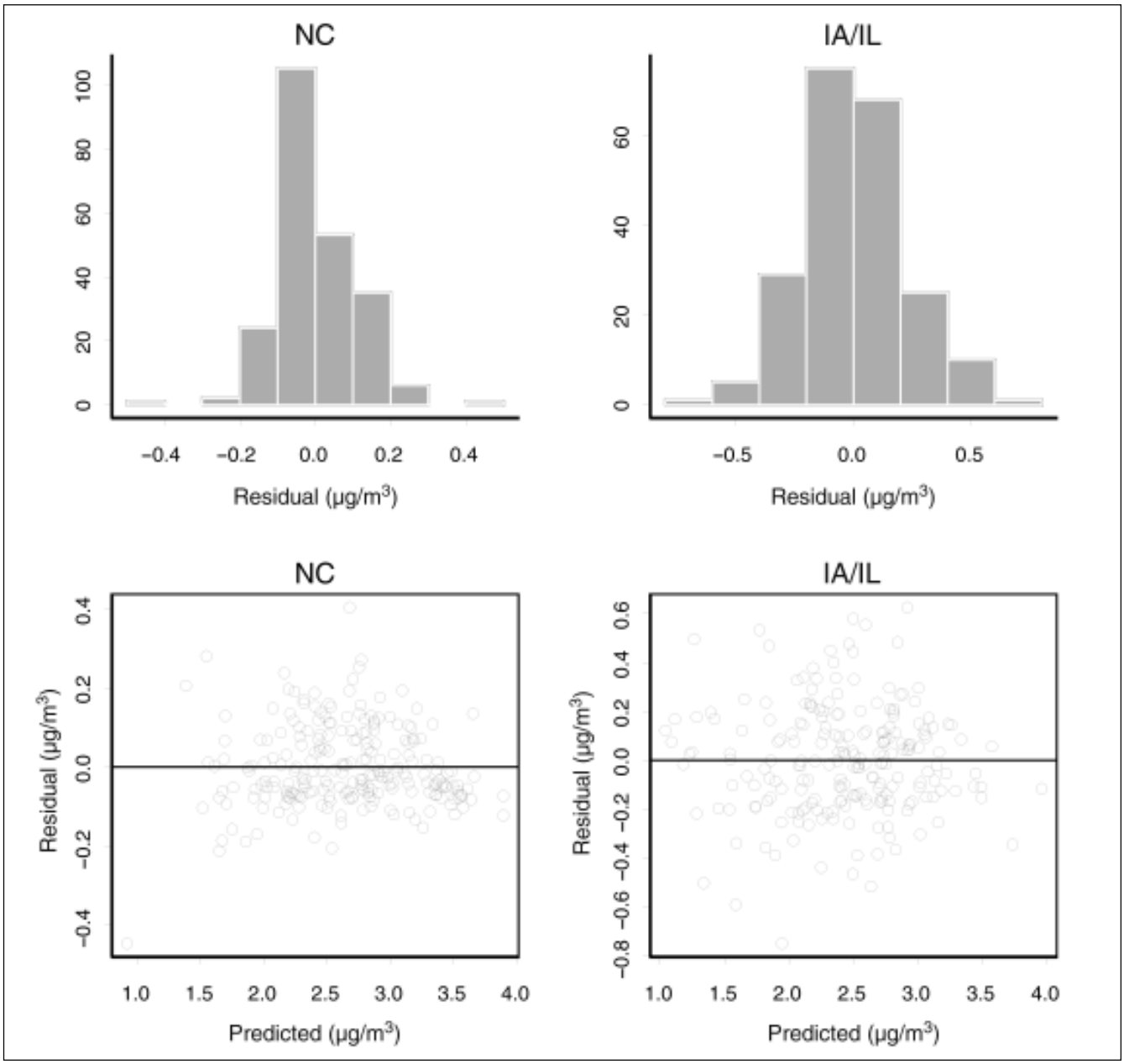


Figure 4. Regression diagnostics for final models in NC and IA/IL.

AQI Reporting

Once an acceptable model is obtained, it may be used along with CM measurements to provide timelier AQI reporting. Because the model relates daily averages, using it to report a rolling 24-hr average AQI would appear to be more appropriate than reporting the most recent hour's worth of data. While a rolling 24-hr average includes 23 hr of past data, it remains a more timely reporting mechanism because it includes the most recent hour's worth of data. This fact is highlighted in Figures 5 and 6, which display 2 weeks of AQI reporting using once-daily FRM measurements versus hourly CM measurements averaged over the most recent 24-hr period. For comparison, the figures display AQIs calculated from both transformed (i.e., model adjusted) and untransformed CM measurements.

To obtain the AQI from modeled FRM measurements, the CM 24-hr averages were first transformed using the appropriate model, summarized in Table 1. The AQI value was then calculated according to the following description.[1] Suppose a CM $PM_{2.5}$ 24-hr average measurement of 20 $\mu g/m^3$ is observed. Then, according to the NC log-linear model of Table 1, for example, the appropriate model-based FRM measurement to use in reporting the AQI is

$$FRM_{model} = \exp\{[-0.114 + 1.054*\ln(20)]\} = \exp\{3.0435\} = 20.98\ \mu g/m^3 \quad (13)$$

Plugging 20.98 $\mu g/m^3$ into the formula for the AQI yields a reported index value of

$$PM_{2.5}: \quad \frac{(100 - 51)}{(40.4 - 15.5)}(20.98 - 15.5) + 51 = 62 \quad (14)$$

The AQI values reported in Figures 5 and 6 were calculated from NC data and IA/IL data, respectively, during the 2-week period of February 13–February 27, 2000. The model adjustment for the IA/IL data is clearly more significant than that for the NC data, indicating a stronger bias between the CM and FRM measurements at the IA/IL monitoring site relative to the NC site. Consider the 15 midnight-to-midnight averaging periods during this 2-week period, and compare the AQI classification (0–50 = Good; 51–100 = Moderate) using FRM measurements to that using either raw CM (untransformed) or model-transformed CM (modeled FRM) measurements. In NC, the resulting AQI classifications match in 14 of 15 cases, regardless of whether raw or model-transformed CM measurements are used. In IA/IL, one of the 15 days is missing

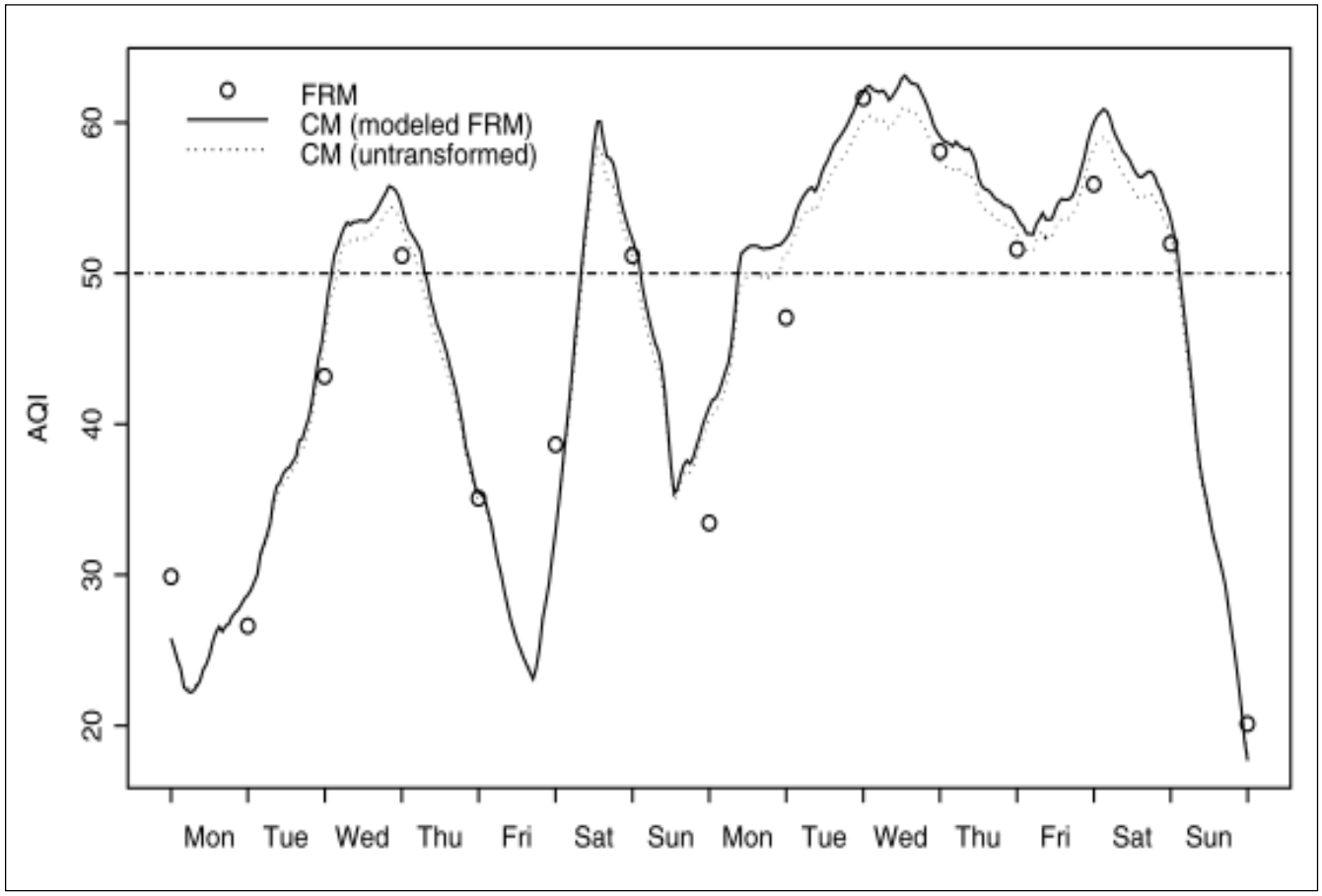


Figure 5. $PM_{2.5}$ AQI values based on CM and FRM measurements in NC from February 13 to February 27, 2000.

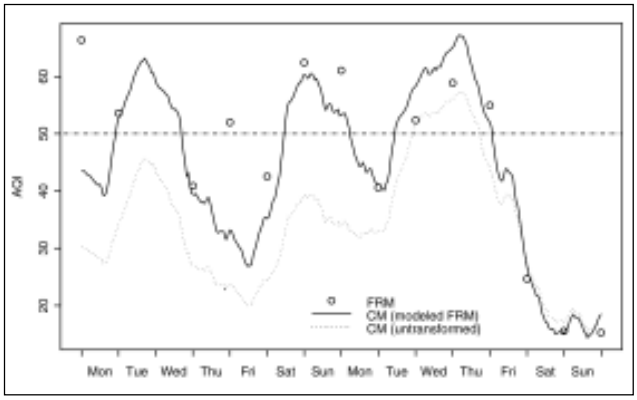


Figure 6. $PM_{2.5}$ AQI values based on CM and FRM measurements in IA/IL from February 13 to February 27, 2000.

the FRM measurement; thus, there are only 14 periods. The resulting AQI classifications match in 12 of 14 cases when model-transformed CM measurements are used, but in only 8 of 14 cases when raw CM measurements are used.

CONCLUSIONS

EPA's DQO process was employed to develop a measure of adequacy for deciding whether a statistical linear regression model that relates FRM and continuous $PM_{2.5}$ measurements is sufficient for use in AQI reporting. Next, MSA-specific models were developed for NC and IA/IL that relate FRM $PM_{2.5}$ measurements with CM measurements. Note that the DQO steps are not specific to an MSA, but the model development is. For this paper, the purpose of the models is to allow for the use of CM measurements in providing timelier reporting of an AQI. However, other applications requiring daily FRM-like $PM_{2.5}$ measurements can benefit from these models. Because the models were based on recent data specific to the time of their development, their validity should be checked on a periodic basis. How often an MSA checks and updates its model will depend on how varied the monitors in its area tend to be. It will probably take at least a quarter's worth of data to make any significant change, unless there are changes to the operating procedures of the CM.

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